1.

[*] HW5_1.cpp

```cpp
#include <iostream>
using namespace std;

long long factorial( int num )
{
    long long fact = 1;
    for ( int i = 1; i <= num; ++i ) fact *= i;
    return fact;
}


int main()
{
    int n,k;

    cout <<"Enter a positive integer for n (1 to 20):";
    cin >> n;
    cout <<"Enter a positive integer for k (k <= n):";
    cin >> k;

    long long ans = factorial(n) / factorial(n - k);

    cout << "Permutation of n object k at a time:" << ans;

    return 0;
}
```

```
Enter a positive integer for n (1 to 20):6
Enter a positive integer for k (k <= n):4
Permutation of n object k at a time:360
--------------------------------
Process exited after 9.561 seconds with return value 0
請按任意鍵繼續 . . .
```

2.

HW5_2.cpp

```cpp
#include <iostream>
using namespace std;

void printPascalTriangle( int n )
{
    int ans[12][12] = {0};
    ans[0][0] = 1;

    for ( int i = 1; i < n + 1; ++i )
    {
        for ( int j = 0; j < i + 1; ++j )
        {
            if ( j == 0 ) ans[i][j] = ans[i-1][j];
            else ans[i][j] = ans[i-1][j-1] + ans[i-1][j];
        }
    }

    cout << "The coefficients are shown below:" << endl;
    for ( int i=0 ; i<n+1 ; ++i )
    {
        for( int j=0 ; j<i+1 ; ++j ) cout << ans[i][j] << " ";
        cout << endl;
    }

}


int main()
{
    int n;

    cout << "Enter the size of the Pascal triangle:";
    cin >> n;

    printPascalTriangle(n);

    return 0;
}
```

```
Enter the size of the Pascal triangle:5
The coefficients are shown below:
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1

--------------------------------
Process exited after 2.481 seconds with return value 0
請按任意鍵繼續 . . .
```

3.

HW5_3.cpp

```cpp
#include <iostream>
#include<iomanip>
using namespace std;

void printEmirpNumbers()
{
    int prime[1000],ans[100]={0};

    for ( int i=0 ; i<1000 ; ++i ) prime[i]=i;

    // Prime
    for ( int i=2 ; i<1000 ; ++i )
    {
        if( prime[i]!=0 )
        {
            for( int j=i*i ; j<1000 ; j+=i ) prime[j]=0;
        }

    }
    // Emirp
    for( int i=2,j=0 ; i<1000 ; ++i )
    {
        if( prime[i]!=0 )
        {
            int temp = i;
            int rev = 0;
            while( temp>0 )
            {
                rev = rev*10 + temp%10;
                temp /= 10;
            }

            if( prime[rev]!=0 )
            {
                ans[j]=i;
                ++j;
            }
        }
    }

    cout << "The emirp numbers less than 1000:" << endl;
    for( int i=0 ; i<100 ; ++i )
    {
        if(ans[i]==0 ) break;
        if( i!=0 && i%10==0 ) cout << endl;
        cout << setw(4) << ans[i];
    }

}


int main()
{
    printEmirpNumbers();

    return 0;
}
```

```
The emirp numbers less than 1000:
   2   3   5   7  11  13  17  31  37  71
  73  79  97 101 107 113 131 149 151 157
 167 179 181 191 199 311 313 337 347 353
 359 373 383 389 701 709 727 733 739 743
 751 757 761 769 787 797 907 919 929 937
 941 953 967 971 983 991
--------------------------------
Process exited after 0.2072 seconds with return value 0
請按任意鍵繼續 . . .
```

4.

[*] HW5_4.cpp

```cpp
#include <iostream>
using namespace std;

bool isDivisible( int a,int b )
{
    if ( a % b == 0 ) return true;
    else return false;
}


int main()
{
    int n;
    cout << "Enter a number between 1 and 100:";
    cin >> n;

    cout << "The number:" << n << endl;
    cout << "The factors:";

    for( int i=1 ; i<n+1 ;++i )
    {
        if( isDivisible(n, i) )
            cout << i << " ";
    }

    return 0;
}
```

```
Enter a number between 1 and 100:60
The number:60
The factors:1 2 3 4 5 6 10 12 15 20 30 60
--------------------------------
Process exited after 1.428 seconds with return value 0
請按任意鍵繼續 . . .
```